



MATHEMATICS (EXTENSION 1)

2012 HSC Course Assessment Task 2

March 12, 2012

General instructions

- Working time – 55 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Weiss

☐ 12M3C – Ms Ziaziaris

☐ 12M4B – Mr Ireland

☐ 12M3D – Mr Lowe

☐ 12M4C – Mr Fletcher

☐ 12M3E – Mr Lam

Marker's use only.

QUESTION	1-6	7	8	9	10	11	12	Total	%
MARKS	$\overline{6}$	$\overline{10}$	$\overline{10}$	$\overline{8}$	$\overline{8}$	$\overline{4}$	$\overline{4}$	$\overline{50}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. If $y = e^{\ln x}$, then $\frac{dy}{dx} =$ **1**
- (A) 1 (C) $\frac{1}{x}$
- (B) 0 (D) none of these
2. If $\log_a b = c$, then **1**
- (A) $a = b^c$ (C) $b = c^a$
- (B) $c = b^a$ (D) $b = a^c$
3. A logarithm is another name for **1**
- (A) a base (C) an operator
- (B) an index (D) none of these
4. If $x^{k+3} = e^{7 \ln x}$, the value of k is **1**
- (A) e (C) -3
- (B) 0 (D) 4
5. $\int \frac{e^x}{1 + e^x} dx$ evaluates to **1**
- (A) $e^x + C$ (C) $\ln(1 + e^x) + C$
- (B) $\ln(1 + e^{2x}) + C$ (D) $\ln(1 - e^x) + C$
6. Given that $\log_8 2 = \log_x 5$, then $x =$ **1**
- (A) $\frac{1}{2}$ (C) 2
- (B) $\frac{1}{3}$ (D) none of the above

Examination continues overleaf...

Section II: Short answer

Question 7 (10 Marks)	Commence a NEW page.	Marks
(a) Draw an accurate sketch of the function $y = \log_e(2x + 1)$, showing its essential features. State its domain and range.		4
(b) Differentiate:		
i. $\log_e \left(\frac{2x + 1}{3x - 7} \right)$.		2
ii. 5^x .		2
iii. $x^3 e^{-3x}$.		2

Question 8 (10 Marks)	Commence a NEW page.	Marks
(a) Write the primitive of		
i. $\frac{x}{9 + x^2}$		2
ii. $\frac{2}{x} + 5e^x$.		2
iii. $\frac{6 - 2x - x^2}{x}$.		2
iv. $x(x^2 + 4)^5$.		2
(b) Evaluate $\int_1^4 y \, dx$ if $xy = 1$.		2

Question 9 (8 Marks)	Commence a NEW page.	Marks
(a) i. Show that $\frac{d}{dx}(x \ln x - x) = \ln x$.		2
ii. Hence or otherwise, find $\int \ln x^2 \, dx$.		2
(b) For the curve $y = \frac{e^x}{x^2 + 1}$, find the stationary point and determine its nature.		4

Question 10 (8 Marks)		Commence a NEW page.	Marks
(a)	The gradient of a curve is given by $\frac{dy}{dx} = \frac{2}{2x-1}$ and the curve passes through $(1, \log_e 3)$. Find the equation of the curve.		3
(b)	i. Evaluate $\int_0^4 (x^2 - 2x) dx$.		2
	ii. Find the area bounded by the curve $y = x^2 - 2x$, the x axis and the ordinates $x = 0$ and $x = 4$.		2
	iii. What do you notice about your answers to parts (i) and (ii)? Give a brief explanation.		1

Question 11 (4 Marks)		Commence a NEW page.	Marks
(i)	Find the volume formed when the portion of the curve $xy = 1$ between $x = 1$ and $x = a$ is rotated about the x axis.		3
(ii)	What is the limiting value of this volume as $a \rightarrow \infty$?		1

Question 12 (4 Marks)		Commence a NEW page.	Marks
Prove by mathematical induction:			4

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \cdots + 2n(2n-1) = \frac{n(n+1)(4n-1)}{3}$$

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

6 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

Section I

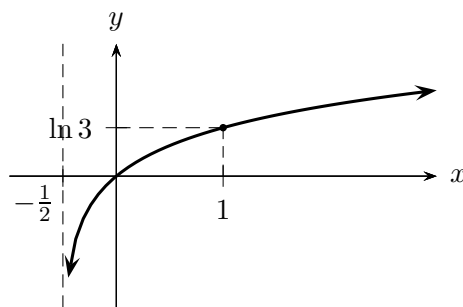
1. (A) 2. (D) 3. (B) 4. (D) 5. (C) 6. (D)

Question 7 (Lowe)

(a) (4 marks)

✓ [2] for graph.

✓ [1] each for domain & range.



$$D = \left\{ x : x \geq -\frac{1}{2} \right\} \quad R = \{ y : y \in \mathbb{R} \}$$

(b) (2 marks)

✓ [1] for using log laws correctly.

✓ [1] for $\frac{2}{2x+1} - \frac{3}{3x-7}$

$$\begin{aligned} & \frac{d}{dx} \left(\log_e \left(\frac{2x+1}{3x-7} \right) \right) \\ &= \frac{d}{dx} (\log_e(2x+1) - \log_e(3x-7)) \\ &= \frac{2}{2x+1} - \frac{3}{3x-7} \left(= \frac{-17}{(2x+1)(3x-7)} \right) \end{aligned}$$

(c) (2 marks)

✓ [-1] for each incorrect step.

$$y = 5^x$$

Take log base e on both sides,

$$\ln y = \ln 5^x = x \ln 5$$

$$\therefore y = e^{x \ln 5}$$

$$\begin{aligned} \frac{d}{dx} (5^x) &= \frac{d}{dx} (e^{x \ln 5}) \\ &= (\ln 5) e^{x \ln 5} \\ &= (5^x \ln 5) \end{aligned}$$

(d) (2 marks)

✓ [1] for correctly applying the product rule.

✓ [1] for correct simplification to $-3x^3 e^{-3x} + 3x^2 e^{-3x}$

$$\begin{aligned} y &= x^3 e^{-3x} \\ u &= x^3 \quad v = e^{-3x} \\ u' &= 3x^2 \quad v' = -3e^{-3x} \\ \frac{d}{dx} (x^3 e^{-3x}) &= x^3 (-3e^{-3x}) + 3x^2 e^{-3x} \\ &= -3x^3 e^{-3x} + 3x^2 e^{-3x} \\ &= (3x^2 e^{-3x} (1-x)) \end{aligned}$$

Question 8 (Ziaziaris)

(a) i. (2 marks)

✓ [-1] for each error.

$$\int \frac{x}{9+x^2} dx = \frac{1}{2} \log_e (9+x^2) + C$$

ii. ✓ [-1] for each error.

$$\int \frac{2}{x} + 5e^x dx = 2 \log_e x + 5e^x + C$$

iii. ✓ [-1] for each error.

$$\begin{aligned} \int \frac{6-2x-x^2}{x} dx &= \int \frac{6}{x} - 2 - x dx \\ &= 6 \log_e x - 2x - \frac{1}{2} x^2 + C \end{aligned}$$

iv. ✓ [-1] for each error.

$$\begin{aligned} \int x (x^2+4)^5 dx &= \frac{1}{2} \int 2x (x^2+4)^5 dx \\ &= \frac{1}{2} \times \frac{1}{6} (x^2+4)^6 + C \\ &= \frac{1}{12} (x^2+4)^6 + C \end{aligned}$$

(b) (2 marks)

✓ [-1] for each error.

$$\int_1^4 \frac{1}{x} dx = \left[\log_e x \right]_1^4$$

$$= \log_e 4 \quad (= 2 \log_e 2)$$

Question 9 (Lam)

(a) i. (2 marks)

✓ [1] for correctly applying the product rule.

✓ [1] for correct simplification.

$$y = x \ln x - x$$

$$u = x \quad v = \ln x$$

$$u' = 1 \quad v' = \frac{1}{x}$$

$$\frac{dy}{dx} = \cancel{x} \times \frac{1}{\cancel{x}} + \ln x - 1$$

$$= \ln x$$

ii. (2 marks)

✓ [1] for correct use of log law.

✓ [1] for correct answer.

✗ [0] for any other attempts to integrate $\ln(x^2)$.

$$\int \ln(x^2) dx = 2 \int \ln x dx$$

$$= 2(x \ln x - x) + C$$

$$= 2x \ln x - 2x + C \quad \text{(b) i. (2 marks)}$$

(b) (4 marks)

✓ [1] for correct use of quotient rule.

✓ [1] for correct testing to discover horizontal point of inflexion.

✓ [1] for each correct coordinate.

$$y = \frac{e^x}{x^2 + 1}$$

$$u = e^x \quad v = x^2 + 1$$

$$u' = e^x \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{e^x(x^2 + 1) - 2xe^x}{(x^2 + 1)^2}$$

$$= \frac{e^x(x^2 - 2x + 1)}{(x^2 + 1)^2}$$

$$= \frac{e^x(x - 1)^2}{(x^2 + 1)^2}$$

x	1^-	1	1^+
$\frac{dy}{dx}$	$+$	0	$+$
y			

Hence $(1, \frac{1}{2}e)$ is a horizontal point of inflexion.**Question 10** (Weiss)

(a) (3 marks)

✓ [1] for correct primitive.

✓ [1] for correct substitution of point.

✓ [1] for final answer.

$$\frac{dy}{dx} = \frac{2}{2x - 1}$$

$$y = \ln(2x - 1) + C$$

When $x = 1$, $y = \ln 3$

$$\ln 3 = \ln(2 - 1) + C$$

$$\therefore C = \ln 3$$

$$\therefore y = \ln(2x - 1) + \ln 3$$

i. (2 marks)

✓ [1] for correct primitive.

✓ [1] for final answer.

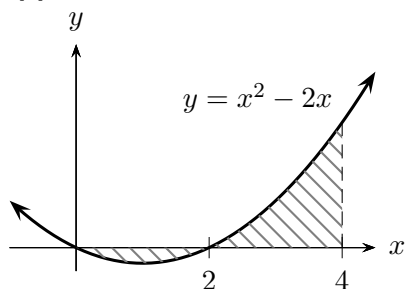
$$\int_0^4 x^2 - 2x dx = \left[\frac{1}{3}x^3 - x^2 \right]_0^4$$

$$= \frac{64}{3} - 16 = \frac{16}{3}$$

ii. (2 marks)

✓ [1] for correct resolution into two separate integrals.

✓ [1] for correct final answer.



$$\begin{aligned}
 A &= \left| \int_0^2 x^2 - 2x \, dx \right| \\
 &\quad + \int_2^4 x^2 - 2x \, dx \\
 &= \left| \left[\frac{1}{3}x^3 - x^2 \right]_0^2 \right| + \left[\frac{1}{3}x^3 - x^2 \right]_2^4 \\
 &= \left| \frac{8}{3} - 4 \right| + \frac{1}{3}(4^3 - 2^3) \\
 &\quad - (4^2 - 2^2) \\
 &= \frac{4}{3} + \frac{56}{3} - 12 \\
 &= 8 \text{ units}^2
 \end{aligned}$$

iii. (1 mark)

Part of the required area is below the x axis.

(ii) (1 mark)

$$\lim_{a \rightarrow \infty} \left[\pi \left(1 - \frac{1}{a} \right) \right] = \pi(1 - 0) = \pi$$

Question 12 (Fletcher)Let $P(n)$ be the statement

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \cdots + 2n(2n-1) = \frac{n(n+1)(4n-1)}{3}$$

Base case $P(1)$:

$$\begin{aligned}
 2(1)(2(1) - 1) &= 2(2 - 1) = 2 \\
 \frac{1(1+1)(4(1) - 1)}{3} &= \frac{2 \times 3}{3} = 2
 \end{aligned}$$

Hence $P(1)$ is true.**Inductive hypothesis:** assume the k -th proposition $P(k)$ is true for some $k \in \mathbb{Z}^+$, i.e.

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \cdots + 2k(2k-1) = \frac{k(k+1)(4k-1)}{3}$$

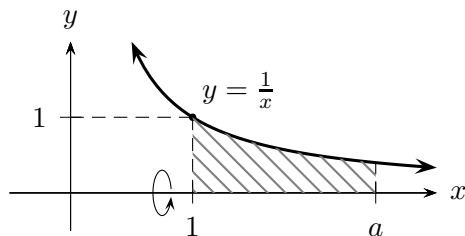
and examine $P(k+1)$:**Question 11** (Ireland)

(i) (3 marks)

✓ [1] for setting up correct integral.

✓ [1] for correct primitive.

✓ [1] for final answer.



$$\begin{aligned}
 V &= \pi \int_1^a \frac{1}{x^2} \, dx \\
 &= \pi \left[-\frac{1}{x} \right]_1^a \\
 &= \pi \left(1 - \frac{1}{a} \right) \text{ units}^3
 \end{aligned}$$

$$\begin{aligned}
 &1 \times 2 + 3 \times 4 + \cdots + 2k(2k-1) \\
 &\quad + 2(k+1)(2(k+1)-1)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{k(k+1)(4k-1)}{3} + 2(k+1)(2k+1) \\
 &= \frac{(k+1)(k(4k-1) + 6(2k+1))}{3} \\
 &= \frac{(k+1)(4k^2 - k + 12k + 6)}{3} \\
 &= \frac{(k+1)(4k^2 + 11k + 6)}{3} \\
 &= \frac{(k+1)(4k+3)(k+2)}{3} \\
 &= \frac{(k+1)((k+1)+1)(4(k+1)-1)}{3}
 \end{aligned}$$

 $\therefore P(k+1)$ is also true. Hence $P(n)$ is true for integers $n \geq 1$ by induction.